

Mathematical Frameworks — Module 03: Perception

— Study Questions

1. Write the two decompositions of variational free energy F .
2. What does it mean to "minimize free energy"? What changes during minimization?
3. What is gradient descent? How is it applied to free energy minimization?
4. Why is the learning rate η important? What happens if it's too large or too small?
5. For a Gaussian model, what does F reduce to? (precision-weighted prediction error)
6. In the equation $F \propto \frac{1}{2} \pi(o - g(s))^2$, identify: o , $g(s)$, π , and $(o - g(s))$.
7. What is precision π ? How does it relate to variance?
8. How does high precision change the influence of a prediction error?
9. How does low precision change the influence of a prediction error?
10. Explain how precision implements mathematical attention.
11. In a hierarchical model, what does each level predict?
12. What flows upward in the hierarchy? What flows downward?
13. How does total free energy relate to the free energies at individual levels?
14. How does prediction error minimization relate to the predictive coding framework in CompNeuro?
15. Can free energy ever reach zero? What would that mean?
16. What is the relationship between free energy and "surprise" (negative log evidence)?
17. How does the accuracy-complexity decomposition explain why perception isn't just pattern-matching?
18. What happens to perception when precision is very high everywhere? Very low everywhere?
19. How does this mathematical framework explain visual illusions?
20. Compare the mathematical description of perception here with the cognitive science (Unit 1) and neuroscience (Unit 2) descriptions.